COPYRIGHT RESERVED BCA (Sem-I) – Programming

Fundamental Using C++ (101) Core-I

2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer all sections as directed.

Section–A

Objective type Questions

Compulsory

1. Choose the correct answer of the following multiple choice questions : 2×10=20

(a) Who developed the C language ?

i) Bjarne Stroustrup

ii) Dennis Ritchie

iii) James Gosling

iv) Ken Thompson

(b) C++ is an extension of which language?

i) Java

ii) Python

iii) C

iv) Pascal

(c) The execution of a C++ program starts from:

i) start()

ii) init()

iii) main()

iv) program()

(d) Which of the following is NOT a basic data type in C++?

i) int

ii) float

iii) char

iv) string

(e) Which keyword is used to declare a constant variable?

i) fixed

ii) const

iii) static

iv) final

(f) Which operator is used for logical AND?

i) &

ii) &&

iii) |

iv) ||

(g) Which C++ object is used for output?

i) cin

ii) cout

iii) printf

iv) scanf

(h) Which statement is used for multiple branching?

i) if

ii) if-else

iii) switch

iv) for

(i) Which loop executes at least once even if the condition is false?

i) for

ii) while

iii) do-while

iv) nested loop

(j) Which function is used to deallocate dynamically allocated memory in C?

i) delete

ii) free()

iii) remove()

iv) dealloc()

Section–B

Short Answer type Questions

Answer any **four** questions of the following : 5×4=20

2. Explain data types in C++.

3. What is an array? Explain one-dimensional array.

4. Explain inheritance in C++.

5. Explain file handling in C++.

6. What is polymorphism? Explain virtual functions.

7. What is exception handling? Explain try, catch and throw.

8. What is a copy constructor? Explain its need.

Section–C

Long Answer type Questions

Answer any **two** questions of the following : 15×2=30

9. Write a C++ program to store student information using structures and display the details.

10. Describe conditional and iterative statements in C++ with examples.

11. Write a C++ program to pass an array to a function and modify its elements.

12. Explain abstract classes and pure virtual functions with examples.

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